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United States Patent	12396901
Kind Code	B2
Date of Patent	August 26, 2025
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Transversely extensible continuous elastic laminate, and a method for producing the same

Abstract

A transversely extensible continuous elastic laminates includes a continuous elastic tape sandwiched between two continuous webs, and at least one row of tabs extending transversely outside the two webs, wherein the tabs are made of a non-woven material and have respective micro-hook formations integrally formed in the non-woven material forming the tabs.

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Appl. No.: 17/696056

Filed: March 16, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20220297417 A1	Sep. 22, 2022

Foreign Application Priority Data

EP	21163679	Mar. 19, 2021
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Publication Classification

Int. Cl.: A61F13/49 (20060101); B32B5/04 (20060101); B32B37/00 (20060101); B32B37/04 (20060101); B32B37/20 (20060101); B32B38/00 (20060101)

U.S. Cl.:

CPC **A61F13/49015** (20130101); **B32B5/04** (20130101); **B32B37/0053** (20130101);
B32B37/0084 (20130101); **B32B37/04** (20130101); **B32B37/206** (20130101);
B32B38/0004 (20130101); **B32B38/1875** (20130101); A61F2013/49022 (20130101);
A61F2013/49033 (20130101); B32B2038/0028 (20130101); B32B2305/20 (20130101);
B32B2310/00 (20130101); B32B2555/02 (20130101)

Field of Classification Search

CPC: A61F (13/49015); A61F (2013/49022); A61F (2013/49033); A61F (13/15756); A61F
(13/622); A61F (13/62); A61F (13/15577); A61F (13/15723); A61F (13/15764); A61F
(13/56); A61F (2013/15821); B32B (5/04); B32B (37/0053); B32B (37/0084); B32B
(37/04); B32B (37/206); B32B (38/0004); B32B (38/1875); B32B (2038/0028); B32B
(2305/20); B32B (2310/00); B32B (2555/02); B32B (3/08); B32B (3/28); B32B (5/022);
B32B (5/26); B32B (27/00); B32B (27/12); B32B (37/00); B32B (2555/00); A44B
(18/0061)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority to European Patent Application No. 21163679.0 filed Mar. 19, 2021. The disclosure of the above application is incorporated herein by reference in its entirety.

FIELD OF THE INVENTION

(2) The present invention relates to a transversely extensible continuous elastic laminate.

(3) The present invention has been developed with particular reference to the production of

disposable absorbent sanitary articles, such as, for example, diapers, training pants, absorbent sanitary products for incontinent adults, etc.

(4) A transversely extensible continuous elastic laminate may constitute a semi-finished intermediate product which can be used for supplying machines for manufacturing absorbent sanitary articles. The transversely extensible continuous elastic laminate can be cut transversely to produce elastic side panels for absorbent sanitary articles. Discrete left and right fastening tabs could be already attached externally on the elastic laminate.

(5) Embodiments of the present invention also relate to an elastic side panel for absorbent sanitary articles.

(6) The present invention also relates to methods for producing transversely extensible continuous elastic laminates.

PRIOR ART

(7) An absorbent sanitary article wearable as a pant typically has a structure that comprises a rectangular-shaped central body or chassis and at least one pair of elastic side panels which extend laterally from opposite sides of the central body in the front or rear section of the central body. The side panels are provided with fastening tabs, typically comprising micro-hooks pads, which can be releasably attached to a front panel for closing the absorbent sanitary article around the waist of the user.

(8) EP-A-3213728 discloses a transversely extensible elastic laminate comprising: a first and a second web, having respective pleated central portions, an elastic tape sandwiched between the pleated central portions of the first and second web and elastically stretchable in a transverse direction, and a plurality of fastening tabs spaced apart from each other in a longitudinal direction.

(9) The fastening tabs each include a non-woven support web of soft material and a micro-hook pad of relatively rigid thermoplastic material fixed on a surface of the non-woven support web by glue or by welding.

(10) The micro-hook pads are produced starting from continuous tapes, usually made of relatively rigid thermoplastic material, having continuous or intermittent micro-hook formations integrally formed thereon. The continuous tape with integrally formed micro-hook formations is cut to form a plurality of discrete micro-hook pads. Such discrete micro-hook pads are spaced by a desired pitch and are fixed in spaced positions on a surface of a continuous non-woven support web. The continuous non-woven support web with the micro-hook pads fixed thereon is then cut to form fastening tabs each including a non-woven support web of soft material and a micro-hook pad of relatively rigid thermoplastic material fixed on a surface of the non-woven support web by glue or by welding.

(11) The continuous tapes with integrally formed micro-hook formations is manufactured by suppliers specialized in manufacturing hook-and-loop fasteners. The micro-hook tapes are packaged in reels which are delivered to the manufacturing plants of absorbent sanitary articles. In the machines for manufacturing absorbent sanitary articles the reels of continuous tapes with integrally formed micro-hook formations are unwound, cut, spaced, and fixed to the continuous non-woven support web. All these operations require complex and expensive dedicated equipment which increase the complexity of the machines for manufacturing absorbent sanitary articles. The cost of the continuous tapes with integrally formed micro-hook formations is high, therefore, the dimensions of the micro-hook pads or patches is kept to the minimum possible, which requires complex equipment for cutting the continuous tapes with integrally formed micro-hook formations in very small pieces and for feeding such small pieces with high precision to the continuous non-woven support web.

(12) A further drawback of the prior art is that the micro-hook fastening pads are made of relatively rigid thermoplastic material. The user usually perceives unpleasant stiffness during the application of the sanitary article to the wearer. It is not unusual that the user gets scratches or small cuts on the soft skin under the nails due to the contact with sharp and rigid edges of the micro-hook pads.

OBJECT AND SUMMARY OF THE INVENTION

(13) The object of the present invention is to provide a transversely extensible continuous elastic laminate usable to produce elastic side panels for absorbent sanitary articles, which overcomes the problem of the prior art.

(14) According to a first aspect of the invention, this object is achieved by a transversely extensible continuous laminate having the features of claim 1.

(15) An important feature of the present invention is that the tabs with micro-hook formations can be manufactured in-line in a machine for manufacturing absorbent sanitary articles starting from reels of non-woven material. Therefore, there is no need to supply the machine for manufacturing absorbent sanitary articles with reels of micro-hook tapes supplied by companies specialized in manufacturing hook-and-loop fasteners.

(16) The present invention eliminates handling and transportation of micro-hook tapes, which reduces costs, simplifies handling of raw material in the plants for manufacturing absorbent sanitary articles, and has a positive impact on sustainability due to the elimination of the need of transporting the micro-hook tapes.

(17) According to another aspect, the present invention relates to an elastic side panel for absorbent sanitary articles.

(18) According to another aspect, the invention relates to methods for producing transversely extensible laminates.

(19) The manufacturing methods, as compared to the prior art, provide a better control on the distance between the fastening tabs, which in the method according to the invention are not applied as discontinuous discrete elements. This reduces manufacturing wastes and has a direct positive impact on sustainability.

(20) The claims form an integral part of the technical disclosure provided here in relation to the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present invention will now be described with reference to the attached drawings, provided purely by way of non-limiting example, wherein:

(2) FIGS. **1-10** are schematic views showing various steps of a first embodiment of a method for producing a transversely extensible continuous elastic laminate having integral micro-hook fastening formations,

(3) FIG. **11** is a schematic plan view of a transversely extensible continuous elastic laminate in an extended position,

(4) FIG. **12** is a schematic cross-section taken along the line XII-XII of FIG. **11**,

(5) FIG. **13** is a schematic plan view showing the transversely extensible continuous elastic laminate of FIG. **11** in a folded configuration,

(6) FIG. **14** is a schematic cross-section taken along the line XIV-XIV of FIG. **13**,

(7) FIG. **15** is a schematic plan view showing an elastic side panel for absorbent sanitary articles in an extended position,

(8) FIG. **16** is a schematic plan view showing an elastic side panel with a different shape,

(9) FIGS. **17-20** are schematic plan views showing various steps of a second embodiment of a method for producing a transversely extensible continuous elastic laminate having integral micro-hook fastening formations,

(10) FIG. **21** is a schematic plan view of a second embodiment of a transversely extensible continuous elastic laminate in an extended position,

(11) FIG. **22** is a schematic cross-section taken along the line XXII-XXII of FIG. **21**,

(12) FIGS. **23-29** are schematic plan views showing various steps of a third embodiment of a method for producing a transversely extensible continuous elastic laminate having integral micro-hook fastening formations,

(13) FIG. **30** is a schematic plan view of a third embodiment of a transversely extensible continuous elastic laminate in an extended position, and

(14) FIG. **31** is a schematic cross-section taken along the line XXXI-XXXI of FIG. **30**.

(15) In the following description, identical or similar components will be indicated by the same reference numerals.

(16) It should be appreciated that the attached drawings are schematic and not to scale with respect to real products. Various figures may not be represented in the same scale. Also, in various figures some elements may not be shown to better show other elements.

DETAILED DESCRIPTION

(17) FIGS. **1-10** schematically show various steps of a method for producing a transversely extensible continuous elastic laminate.

(18) In a first step, shown in FIG. **1**, two first continuous webs **10** are fed in a longitudinal direction X. The two first continuous webs **10** may be made of a non-woven material. In a possible embodiment, the two first continuous webs **10** may be obtained from a single continuous web unwound from a reel and cut longitudinally so as to form two separate webs advancing in the longitudinal direction X.

(19) The two first continuous webs **10** are spaced apart from each other in a transversal direction Y orthogonal to the longitudinal direction X.

(20) Each of the two first continuous webs **10** has a central portion **12** and first and second edge portions **14**, **16** opposite to each other with respect to the central portion **12**.

(21) The two first continuous webs **10** may be positioned on the outer surface of a wheel or on the upper surface of a conveyor belt as they advance continuously in the longitudinal direction X.

(22) In a second step, shown in FIG. **2**, two continuous elastic tapes **18** are fed in the same longitudinal direction X and are overlapped to the respective central portions **12** of the two first continuous webs **10**. The two elastic tapes **18** may be obtained from a single continuous elastic tape unwound from a reel and cut longitudinally so as to form the two separate elastic tapes advancing in the longitudinal direction X.

(23) The two elastic tapes **18** are elastically stretched in the transversal direction Y when they are applied to the respective central portions **12** of the two continuous webs **10**. The two continuous elastic tapes **18** may have the capability to extend in the transversal direction Y by at least 100%, and preferably by 300%, with respect to the respective rest dimensions and to return to the respective rest dimensions in the absence of a transversal force. In a possible embodiment, the degree of transversal elastic stretching of the two continuous elastic tapes **18** may be in the order of 200%. This means that the two continuous elastic tapes **18** are applied on the respective first continuous webs **10** with a width essentially equal to three times the width that the continuous elastic tapes **18** have at rest, i.e., in the absence of transversal forces.

(24) The transverse extension of the two continuous elastic tapes **18** may be obtained by a spreading device comprising, for each continuous elastic tape **18**, two wheels with respective axes inclined with respect to each other, as disclosed in EP-A-3213728.

(25) The transversely stretched continuous elastic tapes **18**, once applied to the respective central portions **12** of the two first continuous webs **10** may be retained in the stretched state by vacuum suction, as disclosed in EP-A-3213728.

(26) In a possible embodiment, the continuous elastic tapes **18** may have different widths in the transverse direction Y and/or may be stretched in said transverse direction Y to reach different widths.

(27) With reference to FIG. **3**, the method comprises providing a continuous support web **20** made of a non-woven material and having a micro-hook formation **22** integrally formed in the non-

woven material forming the support web **20**. The micro-hook formation **22** may be formed in a central portion of a planar surface **24** of the support web **20**, located centrally between two continuous longitudinal edge portions **26**.

(28) The continuous support web **20** with an integrally formed micro-hook formation **22** may be formed as disclosed in detail in WO2010/085492 A1 and as shown schematically in FIG. 4. With reference to FIG. 4, a smooth non-woven web **27** passes through a micro-hook forming unit **28**. The micro-hook forming unit **28** may comprise a molding roller **29** having a plurality of cavities and a ultrasonic horn **31** which compresses the non-woven web **27** against the surface of the molding roller **29**.

(29) The vibrating energy produced by the ultrasonic horn **31** liquefies or fluidifies locally the non-woven material of the non-woven web **27**, which penetrates in a fluid or liquid state in the cavities of the molding roller **29**. The material of the non-woven web **27** which penetrates in the cavities of the molding roller **29** after cooling forms a micro-hook formation **22** integral with the non-woven material. At the exit of the micro-hook forming unit **28** a continuous support web **20** is formed having an integral micro-hook formation **22**.

(30) With reference to FIG. 5, in a possible embodiment, an additional web **33** may be applied on the central portion of the non-woven web **27** before passing the non-woven web **27** through the micro-hook forming unit **28**. The additional web **33** may be a thermoplastic film or a non-woven web of a material different or identical to the material of the non-woven web **27**. After passing through the micro-hook forming unit **28** the additional web **33**, the micro-hook formation **22** and the non-woven web **27** form an integral structure.

(31) In a possible embodiment, the micro-hook forming unit **28** may be a thermomechanical forming unit, in which the material of the non-woven web **27** is locally liquified or fluidified by heating and compression.

(32) In a possible embodiment, the micro-hook formation **22** may have alternating areas in which the micro-hooks are oriented in opposite directions.

(33) With reference to FIG. 6, the continuous support web **20** with integrally formed micro-hook formation **22** is fed in the longitudinal direction X and the two continuous longitudinal edge portions **26** of the support web **20** are overlapped to respective first edge portions **14** of the two first continuous webs **10**. The two continuous longitudinal edge portions **26** of the continuous support web **20** may be retained on the respective edge portions **14** of the two first continuous webs **10** by the same vacuum suction which holds the stretched elastic tapes **18** on the first webs **10**.

(34) As shown in FIG. 7, in a further step two second continuous webs **30** are fed in the longitudinal direction Y, and are overlapped to the respective first continuous webs **10**.

(35) In a possible embodiment, the two second continuous webs **30** may be obtained from a single continuous web unwound from a reel and cut longitudinally so as to form two separate webs advancing in the longitudinal direction X.

(36) The two second continuous webs **30** are spaced from each other in the transverse direction Y and have respective central portions **32** and respective first and second edge portions **34**, **36** on opposite sides of the central portion **32**. The two second continuous webs **30** may have the same width as the first continuous webs **10**. The two second continuous webs **30** are applied over the respective first continuous webs **10** so that the two continuous elastic tapes **18** are sandwiched between the respective central portions **12**, **32** of the first and second webs **10**, **30**. Also, the two continuous longitudinal edge portions **26** of the support web **20** are sandwiched between respective first edge portions **14**, **34** of the two first and second continuous webs **10**, **30**.

(37) Then, as shown in FIG. 8, the two first and second webs **10**, **30**, the two elastic tapes **18** and the edge portions **26** of the support web **20** are fixed to each other by a pattern of spot welds **40**. The spot welds **40** may be formed by ultrasonic welding as disclosed in EP-A-3213728. The pattern of spot welds **40** may have a reinforced area along the longitudinal portion connecting the first edge portions **14**, **34** of the first and second webs to the edge portions **26** of the support web

20, in order to provide an increased joining strength in that area.

(38) The elastic tapes **18** are stretched in the transversal direction Y during the spot welding which joins the elastic tapes **18** to the first and second webs **10**, **30**. The welding spots **40** may form through holes in the elastic tapes **18** in order to provide breathability features to the composite laminate, as disclosed in EP-A-3213728.

(39) With reference to FIG. **9**, the method further comprises a cutting step wherein the support web **20** is cut along a continuous undulated cutting line **42**. The continuous undulated cutting line **42** may be made by a cutting roller having a shaped cutting blade.

(40) As shown in FIG. **10**, in a possible embodiment, the support web **20** may be cut along two continuous undulated cutting lines **42**, **44** which define a continuous undulated strip **46** comprised between the two cutting lines **42**, **44**. The continuous undulated strip **46** is disposed as waste.

(41) After the cut of the support web **20** along the cutting line **42** or cutting lines **42**, **44**, two separate transversely extensible continuous elastic laminates **50** are formed, each of which has the form shown in FIGS. **11** and **12**.

(42) With reference to FIGS. **11** and **12**, the transversely extensible continuous laminate **50** obtained by the previously disclosed method comprises a first and second continuous web **10**, **30** extending along a longitudinal direction X and a continuous elastic tape **18** elastically extensible in a transversal direction Y and sandwiched between the central portions **12**, **32** of the first and second continuous web **10**, **30** and joined thereto by a pattern of spot welds **40**.

(43) When the vacuum suction which holds the elastic tape **18** stretched in the transversal direction Y is released, the elastic tape **18** contracts in the transversal direction Y. The transversal contraction of the elastic tape **18** forms longitudinal pleats **52**, **54** on the central portions **12**, **34** of the first and second continuous web **10**, **30**. The edge portions **14**, **34** and **16**, **36** of the first and second continuous webs **10**, **30** which extend transversally beyond the elastic tape **18** remain non-pleated.

(44) The transversely extensible continuous elastic laminate **50** comprises a support web **20** made of non-woven material, having a continuous longitudinal edge portions **26** which is joined to the first non-pleated edge portions **14**, **34** of the first and second web **10**, **30** by said pattern of spot welds **40**.

(45) The continuous longitudinal edge portion **26** of the support web **20** may be sandwiched between the first edge portions **14**, **34** of the first and second web **10**, **30**. In a possible embodiment, the continuous longitudinal edge portion **26** of the support web **20** may be fixed on the outer surface of one of the non-pleated edge portions **14**, **34** of the first and second web **10**, **30**.

(46) The pattern of spot welds **40** may be reinforced in the area joining the first non-pleated edge portions **14**, **34** of the first and second web **10**, **30** and the continuous edge portion **26** of the support web **20**.

(47) The support web **20** includes a row of tabs **58** which extend from the continuous longitudinal edge portion **26**. The tabs **58** are connected to each other by the continuous edge portion **26** and are spaced apart from each other in the longitudinal direction X. The tabs **58** are made of non-woven material and have respective micro-hook formations **22** integrally formed with the non-woven material forming the tabs **58**.

(48) With reference to FIGS. **13** and **14**, the tabs **58** of the support web **20** may be folded over the second web **30** along a folding line parallel to the longitudinal axis X, so that the micro-hook formations **22** are releasably fastened to corresponding portions of the second web **30**. Folding of the tabs **58** may be carried out before or after releasing the vacuum suction which keeps the elastic tape **18** stretched in the transverse direction Y. When the tabs **58** are folded as shown in FIGS. **10** and **11**, the transversely extensible continuous elastic laminate **50** can be handled more easily without the risk that the micro-hook formations **22** engage with a non-woven web when the transversely extensible continuous elastic laminate **50** is wound in a reel or during the manufacturing process of absorbent sanitary products.

(49) The transversely extensible continuous elastic laminate **50** is used for producing elastic side

panels in machines for manufacturing absorbent sanitary articles. The transversely extensible continuous elastic laminate **50** may be formed in-line with respect to the manufacturing machine. In a possible embodiment, the transversely extensible continuous elastic laminate **50** can be wound in reels which can be stored before being delivered to a manufacturing machine.

(50) In the machine for manufacturing absorbent sanitary products, the transversely extensible continuous elastic laminate **50** is cut along lines generally transversal to the longitudinal axis X for forming individual elastic side panels. The cutting lines which form the individual elastic side panels may have different shapes so as to form elastic side panels with the desired shape. The cuts which originate the individual elastic side panels are typically carried out while the tabs **58** of the transversely extensible continuous elastic laminate **50** are folded as shown in FIGS. **13** and **14**.

(51) An individual elastic side panel obtained from a transversely extensible continuous elastic laminate **50** is shown in FIG. **15** and is indicated by the reference numeral **60**. The elastic side panel **60** is shown in an extended position for a better understanding. FIG. **12**, which is a cross-section taken along the line XII-XII of the transversely extensible continuous elastic laminate **50** of FIG. **11**, is identical to a cross-section taken along the line XII-XII of the individual elastic side panel **60** of FIG. **15**.

(52) With reference to FIGS. **15** and **12**, the elastic side panel **60** comprises a first and a second web **10**, **30** having respective pleated central portions **12**, **34** having respective pleats **52**, **54** parallel to a first direction X, and respective first and second non-pleated edge portions **14**, **34** and **16**, **36** on opposite sides of the pleated central portions **12**, **32**. An elastic tape **18** is sandwiched between the pleated central portions **12**, **32** of the first and second web **10**, **30**. The elastic tape **18** is elastically stretchable in a second direction Y transversal to the first direction X. The elastic tape **18** is joined to the first and second web **10**, **30** by a pattern of spot welds **40**.

(53) The elastic side panel **60** comprises a support web **20** of a non-woven material including an edge portion **26** joined to the first non-pleated edge portions **14**, **34** by said pattern of spot welds **40**. The pattern of spot welds **40** may be reinforced in the area connecting the first non-pleated portions **14**, **34** of the first and second web **10**, **30** to the edge portion **26** of the support web **20**. The edge portion **26** of the support web **30** in the first direction X has the same length as the non-pleated edge portions **14**, **34** of the first and second web **10**, **30**. The support web **20** includes a tab of non-woven material **58** extending from the edge portion **26**. The tab **58** has an integral micro-hook formation **22** integrally formed with the non-woven material forming the tab **58**.

(54) With reference to FIG. **16**, the elastic side panel **60** may have curved flanks **62**, which may be formed by cutting the transversely extensible continuous elastic laminate **50** along substantially V-shaped curved cutting lines. The portion of the transversely extensible continuous elastic laminate **50** comprised between each of the two substantially V-shaped curved cutting lines is disposed as scrap.

(55) FIG. **17-20** show a second embodiment of a method for producing a transversely extensible continuous elastic laminate. The elements corresponding to those previously disclosed are indicated by the same reference numbers.

(56) With reference to FIG. **17**, a first continuous web **10** is fed in a longitudinal direction X. The first continuous web **10** may be made of a non-woven material.

(57) In a second step, shown in FIG. **18**, two continuous elastic tapes **18** parallel to each other and spaced apart in a transversal direction Y are fed in the longitudinal direction X and are overlapped to respective portions of the first continuous web **10**. The two elastic tapes **18** may be obtained from a single continuous elastic tape unwound from a reel and cut longitudinally so as to form the two separate elastic tapes advancing in the longitudinal direction X.

(58) The two continuous elastic tapes **18** are elastically stretched in the transversal direction Y when they are applied to the continuous web **10**. Transverse stretching and retention of the two elastic tapes **18** on the first continuous web **10** are carried out as previously disclosed.

(59) With reference to FIG. **19**, a second continuous web **30** of a non-woven material is fed in the

longitudinal direction Y and is overlapped to the first continuous web **10**.

(60) The second continuous web **30** has a micro-hook formation **22** integrally formed in the non-woven material forming the second continuous web **30**. The micro-hook formation **22** is formed in a central portion of a planar surface **24** of the second continuous web **30** located between the two continuous elastic tapes **18**. The micro-hook formation **22** may be formed as previously disclosed.

(61) The two continuous elastic tapes **18** are sandwiched between respective portions of the first and second webs **10, 30**.

(62) The first and second web **10, 30**, the two elastic tapes **18** are fixed to each other by a pattern of spot welds **40**. The spot welds **40** may also extend through the micro-hook formation **22**. The elastic tapes **18** are stretched in the transversal direction Y during the spot welding which joins the elastic tapes **18** to the first and second web **10, 30**.

(63) With reference to FIG. **20**, the method further comprises a cutting step wherein the first and second webs **10, 30** are cut along a continuous undulated cutting line **42**. The cutting line **42** cuts across the micro-hook formation **22** but does not cut the two elastic tapes **18**.

(64) After the cut of the first and second web **10, 30** along the cutting line **42**, two separate transversely extensible continuous elastic laminates **50** are formed, each of which has the form shown in FIGS. **21** and **22**.

(65) With reference to FIGS. **21** and **22**, the transversely extensible continuous elastic laminate **50** comprises a first and a second continuous web **10, 30** extending along a longitudinal direction X and a continuous elastic tape **18** elastically extensible in a transversal direction Y and sandwiched between pleated central portions **12, 32** of the first and second continuous web **10, 30** and joined thereto by a pattern of spot welds **40**.

(66) The transversely extensible continuous elastic laminate **50** includes a row of tabs **58** integrally formed with the first and second continuous web **10, 30**. The tabs **58** extend in the transverse direction from the pleated central portions **12, 32** and are spaced apart from each other in the longitudinal direction X.

(67) The tabs **58** have respective micro-hook formations **22** integrally formed with the non-woven material forming the second web **30**.

(68) The tabs **58** may be folded over the second web **30** along a folding line parallel to the longitudinal axis X, so that the micro-hook formations **22** are releasably fastened to corresponding portions of the second web **30**.

(69) The transversely extensible continuous elastic laminate **50** may be cut transversely to form individual elastic side panels as previously disclosed. Each elastic side panel has a respective tab **58** integrally formed with the first and/or second non-woven webs **10, 30** and has a micro-hook formation **22** integrally formed with the non-woven material of one of the two non-woven webs. Each tab **58** may be formed by two webs **10, 30** overlapped and joined to each other by a pattern of spot welds **40**. The pattern of spot welds **40** may be reinforced on the tab **58**. This structure makes the tab **58** particularly resistant. In a possible embodiment, each tab **58** may be formed by a single web **10** or **30** of non-woven material.

(70) FIGS. **23-32** show a third embodiment of a method for producing a transversely extensible continuous elastic laminate. The elements corresponding to those previously disclosed are indicated by the same reference numbers.

(71) With reference to FIG. **23** a first continuous web **10** is fed in a longitudinal direction X. The first continuous web **10** may be made of a non-woven material. The first continuous web **10** has a central portion **12** and first and second edge portions **14, 16** opposite to each other with respect to the central portion **12**.

(72) With reference to FIG. **24**, a continuous elastic tape **18** is fed in the same longitudinal direction X and is overlapped to the central portion **12** of the first continuous web **10**. The elastic tape **18** is elastically stretched in a direction Y transversal to the longitudinal direction X when it is applied to the central portion **12** of the first continuous web **10**.

(73) With reference to FIG. 25, the method comprises providing a continuous support web 20 of non-woven material having an integrally formed micro-hook formation 22. The micro-hook formation 22 may be formed in a central portion of a planar surface 24 of the support web 20. The micro-hook formation 22 may be formed as previously disclosed with reference to FIGS. 4 and 5.

(74) With reference to FIG. 26, the method further comprises a cutting step wherein the continuous support web 20 is cut along two continuous undulated cutting lines 42, 44 passing through the micro-hook formation 22. The two continuous undulated cutting lines 42, 44 define a continuous undulated strip 46, shown by hatched lines in FIG. 26, comprised between the two cutting lines 42, 44 which is disposed as waste. In a possible embodiment, the continuous support web 20 may be cut along only one continuous undulated cutting line.

(75) With reference to FIG. 27, after the cut of the support web 20 along the two continuous undulated cutting lines 42, 44 two separate continuous tab chains 66, are formed, which are offset from each other in the longitudinal direction X. The two continuous tab chains 66, comprise respective continuous longitudinal edge portions 26, and respective tabs 58 projecting from the respective continuous longitudinal edge portions 26 in the transversal direction Y. Each tab 58 has a respective micro-hook formation 22 integrally formed on a respective planar surface 24. In each continuous tab chain 66 the respective tabs 58 are spaced apart from each other in the longitudinal direction X.

(76) After the cut of the support web 20 along the two continuous undulated cutting lines 42, 44 the two separate continuous tab chains 66 have the respective tabs 58 facing inward with respect to a longitudinal centre line. After the cut, the two separate continuous tab chains 66 are crossed with each other as they advance in the longitudinal direction, in order to bring the two separate continuous tab chains 66 in a position in which the respective tabs 58 face outward with respect to a longitudinal centre line.

(77) Then, as shown in FIG. 28, the two continuous longitudinal edge portions 26 of the two continuous tab chains 66 are overlapped, respectively, to the first edge portion 14 and to the second edge portion 16 of the first continuous web 10.

(78) In a further step shown in FIG. 29, a second continuous web 30 is fed in the longitudinal direction X and is overlapped to the first continuous web 10.

(79) The second continuous web 30 may have the same width as the first continuous web 10 and may be made of the same material (e.g. a non-woven material). The second continuous web 30 has a central portion 32 and first and second edge portions 34, 36 opposite to each other with respect to the central portion 12.

(80) The second continuous web 30 is applied over the first continuous web 10 so that the continuous elastic tape 18 is sandwiched between the respective central portions 12, 32 of the first and second webs 10, 30. The continuous longitudinal edge portions 26 of the two continuous tab chains 66 may be sandwiched between respective first edge portions 14, 34 of the first and second continuous web 10, 30 or may be applied on the outer surface of the first or second edge portion 14, 34.

(81) Then, the first and second web 10, 30, the elastic tape 18 and the two continuous longitudinal edge portions 26 of the two continuous tab chains 66 are joined to each other by a pattern of spot welds 40.

(82) After the spot-welding step, a finished transversely extensible continuous elastic laminate 50 is obtained.

(83) The tabs 58 of the two continuous tab chains 66 may be folded over the second web 30 along respective folding lines parallel to the longitudinal axis X, so that the micro-hook formations 22 are releasably fastened to corresponding portions of the second web 30.

(84) With reference to FIGS. 30 and 31, when the vacuum suction which holds the elastic tape 18 stretched in the transversal direction Y is released, the elastic tape 18 contracts in the transversal direction Y. The transversal contraction of the elastic tape 18 forms longitudinal pleats 52, 54 on

the central portions **12, 32** of the first and second continuous web **10, 30**. The edge portions **14, 34** and **16, 36** of the first and second continuous webs **10, 30** which extend transversally beyond the elastic tape **18** remain non-pleated.

(85) With reference to FIGS. **30** and **31**, the transversely extensible continuous elastic laminate **50** obtained by the method previously disclosed comprises a first and second continuous web **10, 30** overlapped to each other and extending along a longitudinal direction X. The first and second continuous web **10, 30** have respective pleated central portions **12, 32** having longitudinal pleats **52, 54** and respective first and second non-pleated edge portions **14, 34** and **16, 36**.

(86) A continuous elastic tape **18** elastically extensible in a transverse direction Y is sandwiched between the pleated central portions **12, 32** of the first and second continuous webs **10, 30** and joined thereto by a pattern of spot welds **40**.

(87) The transversely extensible continuous elastic laminate **50** comprises a first and a second tab chain **66** of non-woven material having respective continuous longitudinal edge portions **26** and respective rows of tabs **58** having respective micro-hook formations **22** integrally formed with the non-woven material of the tabs **58**.

(88) In each tab chain **66** the respective tabs **58** are spaced apart from each other in the longitudinal direction X. The tabs **58** of the two tab chains **66** are offset with respect to each other.

(89) The continuous longitudinal edge portions **26** of the first and second tab chains **66** are joined to the respective first and second non-pleated edge portions **14, 34** and **16, 36** of the first and second continuous web **10, 30** by said pattern of spot welds **40**. The pattern of spot welds **40** may be reinforced in the areas joining the continuous longitudinal edge portions **26** of the first and second tab chain **66** and the respective first and second non-pleated edge portions **14, 34** and **16, 36** of the first and second continuous web **10, 30**.

(90) In a possible embodiment, the micro-hook formations **22** of the first and second tab chain **66** may have respective hooks oriented in opposite direction to each other.

(91) The transversely extensible continuous elastic laminate **50** may be cut along cutting lines **70** (FIG. **30**) for forming individual elastic side panels **60**, as previously disclosed.

(92) An absorbent sanitary article may have an absorbent central body and two or four elastic side panels **60** as previously described fixed to two opposite longitudinal edges of said central body.

(93) As compared to the prior art disclosed in EP-A-3213728, the elastic side panels **60** according to the present invention have an increased resistance and are less exposed to the risk of breaks because the tabs **58** are connected to the webs **10, 30** along the whole length of the webs **10, 30**.

(94) In use, the tabs **58** have a soft feeling to the touch because the micro-hook formations **22** are formed by non-woven material which is softer than the relatively rigid thermoplastic material forming the micro-hook pads of the prior art.

(95) The method for manufacturing the transversely extensible continuous elastic laminate **50** does not require the application of discrete tabs on the edges of the elastic laminate. The tabs with the respective micro-hook formations **22** are formed by applying and cutting a continuous web. This involves considerable advantages in that there is no need for complex and expensive repitch devices for applying discrete elements at a constant pitch.

(96) Of course, without prejudice to the principle of the invention, the details of construction and the embodiments can be varied, even significantly, with respect to those illustrated here without departing from the scope of the invention as defined by the following claims.

Claims

1. A method for producing a continuous transversely extensible elastic laminate comprising: feeding in a longitudinal direction two first continuous webs, the two first continuous webs being spaced from each other in a transverse direction and having respective central portions and respective first and second edge portions on opposite sides of the respective central portions,

feeding in said longitudinal direction two continuous elastic tapes, elastically stretching said two continuous elastic tapes in the transverse direction transversal to said longitudinal direction, and applying said two continuous elastic tapes to respective central portions of said two first continuous webs while stretched in said transversal direction, feeding in said longitudinal direction a continuous support web of a non-woven material having two continuous edge portions, the continuous support web having a micro-hook formation integrally formed in the non-woven material forming said continuous support web, applying said two continuous longitudinal edge portions of said continuous support web to respective first edge portions of said two first continuous webs, feeding in said longitudinal direction two second continuous webs spaced from each other in said transverse direction and having respective central portions and respective first and second edge portions on opposite sides of the respective central portions, applying said two second continuous webs to respective first continuous webs and sandwiching said two continuous elastic tapes between respective central portions of said first and second continuous webs, joining to each other said two first and second continuous webs, said two elastic tapes and the edge portions of the continuous support web by a pattern of spot welds while said two elastic tapes are stretched in said transversal direction, and cutting the continuous support web along at least one continuous undulated cutting line passing through said micro-hook formation so as to form two separate continuous transversely extensible elastic laminates.

2. The method of claim 1, wherein said at least one continuous undulated cutting line forms in each of said continuous transversely extensible elastic laminates a row of tabs connected to each other by a respective continuous longitudinal edge portion and spaced apart from each other in said longitudinal direction.

3. The method of claim 1, wherein said micro-hook formation is integrally formed in said non-woven material by locally liquifying or fluidifying the non-woven material and inserting the liquified or fluidified non-woven material into cavities of a molding roller.

4. The method of claim 3, wherein locally liquifying or fluidifying the non-woven material is obtained by subjecting the non-woven material to ultrasonic compression or to thermomechanical compression.

5. A method for producing a continuous transversely extensible elastic laminate comprising: feeding in a longitudinal direction a first continuous web, feeding in said longitudinal direction two continuous elastic tapes parallel to each other and spaced from each other in a transverse direction transversal to said longitudinal direction, elastically stretching said two continuous elastic tapes in the transverse direction, and applying said two continuous elastic tapes to respective portions of said first continuous web while stretched in said transversal direction, feeding a second continuous web of a non-woven material in said longitudinal direction, the second continuous web having a micro-hook formation integrally formed in the non-woven material forming said second continuous web, applying said second continuous web to the first continuous web and sandwiching said two continuous elastic tapes between respective portions of said first and second continuous webs, joining to each other said first and second continuous web and said two elastic tapes by a pattern of spot welds while said two elastic tapes are stretched in said transversal direction, cutting said first and second continuous webs along at least one continuous undulated cutting line passing through said micro-hook formation so as to form two separate continuous transversely extensible elastic laminates.

6. The method of claim 5, wherein said at least one continuous undulated cutting line forms in each of said continuous transversely extensible elastic laminates a row of tabs connected to each other by a respective continuous longitudinal edge portion and spaced apart from each other in said longitudinal direction.

7. The method of claim 5, wherein said micro-hook formation is integrally formed in said non-woven material by locally liquifying or fluidifying the non-woven material and inserting the liquified or fluidified non-woven material into cavities of a molding roller.

8. The method of claim 7, wherein locally liquifying or fluidifying the non-woven material is obtained by subjecting the non-woven material to ultrasonic compression or to thermomechanical compression.
